DIAGNOSTIC · FIRST · 1/10

Automation can't rush diagnosis.

The debugging order that saves weeks — and why the fix is never the first step.

10 slides · swipe

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'Let's just try it' is gambling.

Symptoms scream. The obvious move looks obvious. But every unverified 'try' teaches the system nothing — and sometimes teaches it a new way to break.

**Random changes teach random lessons,
never the real one.**

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The order I force myself to follow.

01 State the symptom exactly — one sentence, no adjectives

↓

02 List what changed — deploys, config, data, traffic

↓

03 Find the FIRST failing component — not the loudest

↓

04 Only then: change one thing, verify, record

Automation comes after understanding. Always.

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Case study: the printer that wouldn't print.

Klipper hung at boot. Observers restarted it. Same failure. The actual cause: a stale RP2040 config married to a rewired MCU — visible only by reading the config, not by rebooting.

Two reboots. Zero learning.
One config read. One fix.

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Two debugging maps.

GUT

Follow the itch

poke the loudest thing
hope it moves
learn nothing

METHOD

Follow the chain

symptom → boundary
boundary → first fail
first fail → fix + verify

Gut gets lucky. Method gets answers.

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Five questions before touching anything.

01 What exactly broke — symptom, user, time?

02 What changed in the last hour / deploy / config?

03 Where is the first divergence in the logs?

04 What hypothesis dies fastest if I'm wrong?

05 What's the smallest reversible change to test it?

Answer all five and the fix often writes itself.

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The 15-minute rule.

If you can't state the mechanism after 15 minutes of diagnosis, stop changing things. Pull someone in, draw the map, ask the dumb questions — before the smart fix becomes the wrong fix.

Stuck is data.
Acting while stuck is damage.

Why diagnosis-first wins.

The best fix is 90% diagnosis and 10% typing — and everyone wants to skip to the 10.

Skip the 90 and the 10 becomes your new incident.

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The diagnosis law.

Before any automation: name the failure mode, the boundary where it starts, and the signal that says it stopped. Then automate the fix — never the guess.

**Automate consequences,
not guesses.**

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DIAGNOSTIC · KEEP THIS · 10/10

Save this for your next 2 a.m. page.

At 2 a.m. you won't think clearly. Print this order. Symptom, change, first-fail, one-move, verify. Even on zero sleep.

Save for later review

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